UCLA
Department of Statistics

STAT 200B: Theoretical Statistics

Problem Set 7
Winter 2020

Issued: Tuesday, March 3 **Due:** Friday, March 13

Reading: Chapter 8 (except Section 8.7), Chapter 9 (Sections 9.1-9.5 and 9.7)

Full credit: 110 points (out of 240) point.

Problem 7.1 (20 points)

Let p and q be probability densities w.r.t. the d-dimensional Lebesgue measure, and let $f : \mathbb{R}^d \to \mathbb{R}$. We would like to compute

$$\mu_f := \mathbb{E}_p[f(X)], \quad \text{where} \quad X \sim p.$$

Here, $\mathbb{E}_p$ denotes the expectation under density p for X. Let $w(X) := p(X)/q(X)$ be the likelihood ratio based on the two densities and recall the change of measure formula:

$$\mathbb{E}_p[f(X)] = \mathbb{E}_q[w(X)f(X)].$$

The idea of *importance sampling* is to generate an i.i.d. sample $X_1, \dots, X_n \sim q$ and approximate μ_f with

$$\widehat{\mu}_f^{(n)} := \frac{1}{n}\sum_{i=1}^{n} w(X_i)f(X_i),$$

Here, $w(X_i)$ are referred to as importance weights. A common case is where the density $p(\cdot)$ and $q(\cdot)$ are only specified up to constants. In this case, the weights are only specified up to a constant, say $w(X) = Cw_0(X)$ where $w_0(\cdot)$ can be computed. Then, we consider

$$\widetilde{\mu}_f^{(n)} := \frac{\sum_{i=1}^n w(X_i)f(X_i)}{\sum_{i=1}^n w(X_i)} = \frac{\sum_{i=1}^n w_0(X_i)f(X_i)}{\sum_{i=1}^n w_0(X_i)}.$$

Answer the following, assuming that any needed moment is finite:

(a) Show that both $\widehat{\mu}_f^{(n)}$ and $\widetilde{\mu}_f^{(n)}$ are consistent for μ_f.

(b) Derive asymptotic distributions for $\sqrt{n}(\widehat{\mu}_f^{(n)} - \mu_f)$ and $\sqrt{n}(\widetilde{\mu}_f^{(n)} - \mu_f)$.

Problem 7.2 (15 points)

Keener 8.23.

23. Suppose $X_1, \dots, X_n$ are i.i.d. Poisson variables with mean λ and we are interested in estimating $p = P_\lambda(X_i = 0) = e^{-\lambda}$.
    a) One estimator for p is the proportion of zeros in the sample, $\tilde{p} = \#\{i \le n : X_i = 0\}/n$. Find the limiting distribution for $\sqrt{n}(\tilde{p} - p)$.
    b) Another estimator would be the maximum likelihood estimator $\hat{p}$. Give a formula for $\hat{p}$ and determine the limiting distribution for $\sqrt{n}(\hat{p} - p)$.
    c) Find the asymptotic relative efficiency of $\tilde{p}$ with respect to $\hat{p}$.

Problem 7.3 (15 points)

Assume that $X_n \sim \text{Beta}(n, n)$. Use the delta method to show that

$$2\sqrt{2n}\Big(X_n - \frac{1}{2}\Big) \xrightarrow{d} N(0,1).$$

Hint: Recall that X_n can be written as $X_n = \frac{Y_n}{Y_n+Z_n}$ where Y_n and Z_n are independent Gamma$(n,1)$ random variables. Each of Y_n and Z_n can be written as a sum of exponential random variables. Use the multivariate CLT to derive the asymptotic distribution of $[X_n \;\; Y_n]^T \in \mathbb{R}^2$. Then use the delta method.

Problem 7.4 (15 points)

Keener 8.26.

26. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with mean λ, and consider estimating

$$g(\lambda) = P_\lambda(X_i = 1) = \lambda e^{-\lambda}.$$

One natural estimator might be the proportion of ones in the sample:

$$\hat{p}_n = \frac{1}{n}\#\{i \le n : X_i = 1\}.$$

Another choice would be the maximum likelihood estimator, $g(\overline{X}_n)$, with $\overline{X}_n$ the sample average.

a) Find the asymptotic relative efficiency of $\hat{p}_n$ with respect to $g(\overline{X}_n)$.

b) Determine the limiting distribution of

$$n\big[g(\overline{X}_n) - 1/e\big]$$

Problem 7.5 (10 points)

Keener 8.27.

27. Let $X_1, \ldots, X_n$ be i.i.d. from $N(\theta, 1)$, and let $U_1, \ldots, U_n$ be i.i.d. from a uniform distribution on $(0,1)$, with all $2n$ variables independent. Define $Y_i = X_i U_i$, $i = 1, \ldots, n$. If the X_i and U_i are both observed, then $\overline{X}$ would be a natural estimator for θ. If only the products $Y_1, \ldots, Y_n$ are observed, then $2\overline{Y}$ may be a reasonable estimator. Determine the asymptotic relative efficiency of $2\overline{Y}$ with respect to $\overline{X}$.

Problem 7.6 (20 points)

Find a variance-stabilizing transformation for the following: The sample mean $\bar{X}_n$ for i.i.d. samples from Ber(θ). (Here we want the transformed statistic to have asymptotic variance independent of θ.)

Problem 7.7 (50 points)

(5+10+30+5) Consider a linear regression model with random design:

$$y_i = \beta_*^T z_i + w_i, \quad z_i \overset{\text{iid}}{\sim} N(0, I_d), \quad w_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

for $i = 1, \dots, n$, where z_i and w_i are also independent. Here, $\beta_* \in \mathbb{R}^d$ is the "true" regression coefficient. We observe (y_i, z_i) and would like to estimate β_*. Define the *ridge regression estimator* as

$$\widehat{\beta}_n = \operatorname*{arg\,min}_{\beta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^{n} (y_i - \beta^T z_i)^2 + \lambda \|\beta\|_2^2 \tag{1}$$

where $\|\beta\|_2^2 = \sum_{i=1}^d \beta_i^2$ and $\lambda > 0$ is a fixed deterministic number.

(a) Show that the objective function in (1) is a strictly convex function of β. Conclude that the optimization problem has a unique solution $\widehat{\beta}_n$. (The function is in fact strongly convex.)

(b) Show that $\widehat{\beta}_n \xrightarrow{p} \beta_0$ as $n \to \infty$, for a constant vector $\beta_0 \in \mathbb{R}^n$ that you specify.

(c) Show that $\sqrt{n}(\widehat{\beta}_n - \beta_0) \xrightarrow{d} N(0, \Sigma_0)$ for a covariance matrix $\Sigma_0 \in \mathbb{R}^{d \times d}$ that you specify.

(d) Adjust the estimator so that the new estimator is asymptotically consistent for estimating β_* (i.e., it converges to β_* in probability.)

Hint: Assume the conditions we discussed for the consistency and asymptotic normality of M-estimators hold and use those results. Note that β_0 is not equal to β_*.

Problem 7.8 (20 points)

Suppose that $X_1, \dots X_n$ are i.i.d from the normal location model $N(\theta, 1)$, and that we wish to estimate the critical or cutoff value $g_a(\theta) = \mathbb{P}[X_1 \le a]$, where $a \in \mathbb{R}$ is some fixed number.

(a) Let Φ denote the CDF of the standard normal distribution. Show that the estimator

$$\delta_n(X) \quad = \quad \Phi\left(\sqrt{\frac{n}{n-1}}\,(a - \bar{X}_n)\right)$$

is an unbiased estimator of $g_a(\theta)$. Prove that $\sqrt{n}(\delta_n - g_a(\theta)) \xrightarrow{d} N(0, \phi^2(a - \theta))$, where ϕ is the PDF of the standard normal.

(b) If we had doubts about the assumption of normality, we might prefer to use the non-parametric estimator $\delta_n'(X) = \frac{1}{n} \# \{X_i \le a\}$, that simply counts the fraction of X_i that are less than or equal to a. Prove that

$$\sqrt{n}(\delta_n' - g_a(\theta)) \xrightarrow{d} N\left(0, \Phi(a - \theta)[1 - \Phi(a - \theta)]\right).$$

(c) Compute the ARE of δ and δ'. If X_i truly are Gaussian, how much is lost by the non-parametric estimator δ'?

Hint: For showing unbiasedness in part (a) write $\delta_n(X) = \mathbb{P}(Z \leq \cdots \mid X_1, \ldots, X_n)$ where Z is independent of $X_1, \ldots, X_n$ and has a proper distribution. For part (b), write δ_n' as an average of indicator variables.

Problem 7.9 (5 points)

Suppose that for each $\theta_0 \in \Theta$, the set $A(\theta_0)$ is the acceptance region of a level α test for $H_0 : \theta = \theta_0$. For each sample point x, define $S(x) = \{\theta \in \Theta \mid x \in A(\theta)\}$. Show that the random set $S(X)$ is a confidence set of level $1 - \alpha$.

Problem 7.10 (30 points)

Suppose that $X_1, \ldots, X_n$ are independent exponential random variables with $\mathbb{E}[X_i] = \beta t_i$ where $t_1, \ldots, t_n$ are known constants, and $\beta > 0$ is an unknown parameter.

(a) Show that the MLE of β is given by $\widehat{\beta}_n = \frac{1}{n}\sum_{i=1}^n X_i/t_i$.

(b) Prove that $\sqrt{n}(\beta - \widehat{\beta}_n) \xrightarrow{d} N(0, \beta^2)$.

(c) Suppose that we want to test $H_0 : \beta = 1$ versus $H_1 : \beta \neq 1$. In order to do so, we consider the statistic

$$G(X_1, \ldots, X_n) = \log p(X; \widehat{\beta}_n) - \log p(X; 1),$$

where $p(X; \beta) = \prod_{i=1}^n p(X_i; \beta, t_i)$ is the likelihood. Show that $G(X) = n(\widehat{\beta}_n - \log \widehat{\beta}_n - 1)$.

(d) Show that when H_0 is true, we have $2G(X) \xrightarrow{d} \chi_1^2$.

Problem 7.11 (20 points)

For each of the following problems, compute the generalized likelihood ratio test, and compute its asymptotic distribution under the specified hypothesis H_0.

(a) Let $X_1, \ldots, X_n$ be an i.i.d. sample of $N(\mu_x, \sigma_x^2)$ variates, and let $Y_1, \ldots Y_n$ be an i.i.d. sample of $N(\mu_y, \sigma_y^2)$ variates. Consider testing $H_0 : \mu_x = \mu_y$ and $\sigma_x^2 = \sigma_y^2$.

(b) For $i = 1, \ldots, k$, let $X_{i1}, \ldots, X_{in}$ be independent samples from Poisson distributions with means θ_i, respectively. Consider testing $H_0 : \theta_1 = \theta_2 = \cdots = \theta_k$.

(c) Let $X_1, \ldots, X_n$ be an i.i.d. sample from the exponential distribution with parameter θ, and $Y_1, \ldots Y_n$ an i.i.d sample from the exponential distribution with parameter μ. Consider testing $H_0 : \mu = 2\theta$.

Problem 7.12 (20 points)

A random sample of n individuals are classified into three groups, with probabilities θ^2, $2\theta(1-\theta)$ and $(1-\theta)^2$ respectively, yielding the trinomial distribution over the group sizes (y_1, y_2, y_3):

$$\mathbb{P}[Y_1 = y_1, Y_2 = y_2, Y_3 = y_3] = \frac{n!}{y_1! y_2! y_3!}\theta^{2y_1}\left[2\theta(1-\theta)\right]^{y_2}(1-\theta)^{2y_3}$$

for $y_i \geq 0$ and $y_1 + y_2 + y_3 = n$. (This is known as the *Hardy-Weinburg* model in genetics.)

(a) Find the MLE of θ, and determine the asymptotic distribution $\sqrt{n}(\hat{\theta}_n - \theta)$.

(b) Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$. Suppose that for some k, we have $\mathbb{P}_{\theta_0}[2Y_1 + Y_2 \geq k] = \alpha$. Show that the test that rejects H_0 when $2Y_1 + Y_2 \geq k$ is a UMP level α test for H_0 versus H_1.

(c) Suppose that n is large, and $\alpha = 0.05$. Use asymptotic theory to find an approximate value for k in the UMP test from (b).